

Markov chain Monte Carlo, part 2

Covered by chapter 11 in the textbook and these notes.

Recall, we wish to draw random numbers from $\pi(\boldsymbol{\theta})$, and have only available some function (density kernel) $g(\boldsymbol{\theta}) \propto \pi(\boldsymbol{\theta})$.

The RWMH sampler considered so far works well in low to moderate dimensions, and for target distributions that are not too far from being multivariate normal. For problems not conforming to these assumptions, other alternatives must be considered.

Inefficiency in RWMH sampler arise from the "blind" proposal mechanism, where shape of the target distribution is not taken into account when proposing new states.

Idea: look for a MCMC method where the proposal mechanism is better adapted to the target.

The Metropolis Hastings algorithm

Suppose the proposal mechanism has density $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$, and an initial state $\boldsymbol{\theta}_0$ is given.

Then a generalization of the RWMH algorithm, simply the Metropolis Hastings (MH) algorithm is given by the follows

1. set $t \leftarrow 1$
2. simulate $\boldsymbol{\theta}^* \sim q(\cdot|\boldsymbol{\theta}_{t-1})$.
3. calculate accept probability

$$\alpha = \min \left(1, \frac{q(\boldsymbol{\theta}_{t-1}|\boldsymbol{\theta}^*)g(\boldsymbol{\theta}^*)}{q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})g(\boldsymbol{\theta}_{t-1})} \right)$$

4. simulate $u \sim \text{uniform}(0,1)$.
5. if $u < \alpha$ set $\boldsymbol{\theta}_t = \boldsymbol{\theta}^*$. Otherwise set $\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1}$.
6. set $t \leftarrow t + 1$ and go back to step 2.

Under weak regularity conditions, this process will have stationary distribution $\pi \propto g$.

Some comments:

$$\alpha = \min \left(1, \frac{q(\boldsymbol{\theta}_{t-1}|\boldsymbol{\theta}^*)g(\boldsymbol{\theta}^*)}{q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})g(\boldsymbol{\theta}_{t-1})} \right)$$

Notice: the proposal mechanism density appears in the accept probability.

Notice: $q(\boldsymbol{\theta}_{t-1}|\boldsymbol{\theta}^*)$ = "probability density of moving backwards in time (from $\boldsymbol{\theta}^*$ to $\boldsymbol{\theta}_{t-1}$)"

Notice: $q(\boldsymbol{\theta}_{t-1}|\boldsymbol{\theta}^*) = q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$ for RWMH - just a special case.

Way forward: look for good choices of proposal mechanism $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$

The independent MH sampler

(or independence sampler) obtains from the MH algorithm if we let $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$ be independent of $\boldsymbol{\theta}_{t-1}$, i.e. $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1}) = r(\boldsymbol{\theta}^*)$ for some PDF $r(\boldsymbol{\theta}^*)$.

In this case the accept probability reduces to

$$\alpha = \min \left(1, \frac{r(\boldsymbol{\theta}_{t-1})g(\boldsymbol{\theta}^*)}{r(\boldsymbol{\theta}^*)g(\boldsymbol{\theta}_{t-1})} \right) = \min \left(1, \frac{\frac{g(\boldsymbol{\theta}^*)}{r(\boldsymbol{\theta}^*)}}{\frac{g(\boldsymbol{\theta}_{t-1})}{r(\boldsymbol{\theta}_{t-1})}} \right)$$

For the importance sampler, one wishes as high as possible acceptance rates (as the proposal is independent of the current state).

The independent MH sampler

If r is a good approximation to target distribution π , we expect that $g(\boldsymbol{\theta})/r(\boldsymbol{\theta}) = C\pi(\boldsymbol{\theta})/r(\boldsymbol{\theta}) \approx C$ for all (reasonable) values of $\boldsymbol{\theta}$. Further, in this case, $\alpha \approx 1$.

I.e. the same considerations (and techniques) for choosing the proposal distribution within importance sampling applies also when choosing r .

However, the area of application is different
(importance sampling: integrals, independent
MH: distribution from density kernels)

See R-code `mcmc_2.R`

Metropolis-adjusted Langevin Algorithm (MALA)

Another, much used version of the MH algorithm is MALA. Rather than pre-selecting a fixed proposal distribution, the MALA adapts to the target distribution using the gradient of the log-target density.

First a few steps backs: Consider the stochastic process

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \frac{\Delta}{2} \nabla \log g(\boldsymbol{\theta}_t) + \sqrt{\Delta} \mathbf{z}_t$$

where $\Delta \ll 1$ and $\mathbf{z}_t \sim \text{iid } N(0, \mathbf{I})$. Moreover

$$\nabla \log g(\boldsymbol{\theta}) = \begin{pmatrix} \frac{\partial \log g(\boldsymbol{\theta})}{\partial \theta_1} \\ \vdots \\ \frac{\partial \log g(\boldsymbol{\theta})}{\partial \theta_d} \end{pmatrix}$$

is the gradient of $\log g(\boldsymbol{\theta})$. Letting $\Delta \rightarrow 0$ we obtain continuous time "Langevin diffusion" process. It can be shown that the **Langevin diffusion has $\pi \propto g$ as the stationary distribution.**

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \frac{\Delta}{2} \nabla \log g(\boldsymbol{\theta}_t) + \sqrt{\Delta} \mathbf{z}_t. \quad (1)$$

I.e. in the limit, we know how to construct a process with the desired stationary distribution.

So who needs MCMC? The problem lies in that it is impossible/inefficient to simulate the continuous time limit without introducing errors for a general g .

See R-code `mcmc_2.R`

Rather than to work with "tiny" Δ and many (costly) time steps, we use the representation (1) with larger Δ as the proposal mechanism of the MH algorithm:

$$q(\boldsymbol{\theta}^* | \boldsymbol{\theta}_{t-1}) = \mathcal{N} \left(\boldsymbol{\theta}^* | \boldsymbol{\theta}_{t-1} + \frac{\Delta}{2} \nabla \log g(\boldsymbol{\theta}_{t-1}), \Delta \mathbf{I} \right)$$

Notice; proposals on average towards regions with higher probability density (as opposed to the "blind" proposals of RWMH)

Tuning the MALA

$$q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1}) = \mathcal{N}\left(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1} + \frac{\Delta}{2}\nabla \log g(\boldsymbol{\theta}_{t-1}), \Delta\mathbf{I}\right)$$

Δ used to control behaviour of the process:

”Small” Δ : converges to RWMH with small Σ (high accept rate, but slow exploration).

”Large” Δ : Proposals tend to be too far from current state (low accept rate).

Again, look for some intermediate resolution of the tradeoff.

Higher accept rates (around 50-90%) are desired.

See R-code `mcmc_2.R`

Overview

The methods we have introduced works well in general

- in low dimension (all).
- IMH in high dimension if we have good advance knowledge.
- MALA in moderate dimension if the target is not far from Gaussian.

We will consider one more type of MCMC algorithms : the Gibbs sampler, in next set of slides

Current state of the art: Hamiltonian Monte Carlo (HMC), used in the best available software Stan. No time to cover this unfortunately.

Handling and checking MCMC output

After running an MCMC algorithm and having obtained MCMC samples $\{\boldsymbol{\theta}_t\}_{t=1}^T$, here are some questions you should consider:

- Should the initial samples be discarded (burn-in/warmup) in order to "forget" the initial configuration of the process. If so, how many samples should be discarded?
- How many (post burn-in) samples are required?
- Has the process really reached and explored the complete stationary distribution? (this cannot be known in general, but test can give hints)

Burn-in/warmup

Easy: check trace plots: See R-code `mcmc_2.R`

How many samples do you need? (Assuming burn-in samples have been discarded)

We have already encountered the Effective Sample Size (ESS) in the exercises.

Roughly speaking:

$\text{ESS}(\{\boldsymbol{\theta}_t\}_{t=1}^T) \approx$ How many iid samples from $\pi(\boldsymbol{\theta})$ would you need, so that the average of the iid samples has the same variance as $T^{-1} \sum_{t=1}^T \boldsymbol{\theta}_t$.

Hence $\text{ESS}(\{\boldsymbol{\theta}_t\}_{t=1}^T) = T$ if we have a "perfect" MCMC algorithm producing iid samples.

In practice $\text{ESS}(\{\boldsymbol{\theta}_t\}_{t=1}^T) < T$ for the algorithms considered here.

Very low ESS indicate problems with exploring the stationary distribution, e.g. $\text{ESS}(\{\boldsymbol{\theta}_t\}_{t=1}^T)/T < 0.01$ **usually** indicate trouble.

Confidence interval for $E(\boldsymbol{\theta})$:

$$\bar{\boldsymbol{\theta}}_t \pm z_{\alpha/2} SD(\boldsymbol{\theta}_t) / \sqrt{\text{ESS}(\{\boldsymbol{\theta}_t\}_{t=1}^T)}.$$

The ESS may be estimated from MCMC output (see exercise set 5) See R-code `mcmc_2.R`

How large ESS is needed depends very much on the application, but as a rule of thumb you should always have an $\text{ESS} > 200$.

If your ESS is too small, consider either running more MCMC simulations, or revising your MCMC algorithm.

Note that the ESS refers to the estimation of $E(\boldsymbol{\theta})$. If you are estimating some other quantity, say $E(s(\boldsymbol{\theta}))$, then transform your samples with s and calculate the ESS of $\{s(\boldsymbol{\theta}_t)\}_{t=1}^T$.

See R-code `mcmc_2.R`

Diagnostics

Has the process really reached and explored the complete stationary distribution? **This cannot really be known in general**, but tests can give hints of the opposite.

Geweke test

Based on a single chain.

Checks, using a rather standard two-sample test, whether the first part of the chain and the last part of the chain have the same mean.

(`coda::geweke.test` output: a Z statistic that is standard normal for a stationary chain)

Gelman-Rubin test

Needs multiple chains.

Checks if variance within chain is the same as variance across chains (see book for details.)

(`coda::gelman.test` output: "Potential scale reduction factor", convergence if this quantity is in $[1, 1.1)$)

See R-code `mcmc_2.R`